

SCENARIO-BASED OPTIMIZATION OF A MULTI-STAGE COLD CHAIN NETWORK FOR MOTHER DAIRY

SUPPLY CHAIN ANALYTICS PROJECT

Submitted by
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1. INTRODUCTION

The distribution of perishable food products such as milk and fresh fruits and vegetables require a highly reliable and efficient cold chain network. In India, this challenge is amplified by large geographic distances, demand variability across regions, infrastructure constraints, and the need to protect product quality while maintaining affordability. Mother Dairy Fruit & Vegetable Private Limited, one of India's most prominent dairy and food companies, plays a critical role in supplying essential food products to millions of consumers across the country.

With increasing urbanization, changing consumption patterns, and government support for farmer-centric supply chains, Mother Dairy faces the strategic challenge of expanding its distribution network while ensuring operational feasibility, cost efficiency, and resilience to disruptions. At the same time, improvements in cold chain capability raise an important strategic question: whether enhanced logistics efficiency can enable Mother Dairy to explore selective export opportunities for value-added dairy products without compromising domestic food security.

This case study applies prescriptive analytics using Python-based optimization and simulation to design and evaluate Mother Dairy's cold chain network. The study focuses on greenfield distribution centre (DC) analysis, network optimization, simulation under uncertainty, and risk analysis, with exporting considered as a secondary revenue-enhancing objective enabled by improved feasibility rather than a primary goal.

2. OVERVIEW OF MOTHER DAIRY'S SUPPLY CHAIN

Mother Dairy operates a multi-stage cold chain network comprising milk procurement from farmers, processing plants, cold storage facilities, distribution centres, and regional markets. The company primarily serves domestic demand, with a strong presence in North India and growing operations in Central, Western, and Southern regions.

The supply chain involves two major product categories:

- Milk and milk-based products, which are highly perishable and require strict temperature and time control.
- Fruits and vegetables (F&V), which have moderate perishability and slightly higher tolerance to transport time under refrigeration.

Ensuring uninterrupted cold chain integrity across these products is critical for maintaining quality, safety, and consumer trust.

3. FACILITY LOCATIONS CONSIDERED IN THE STUDY

3.1 Existing Processing Plants (Present)

- Motihari, Bihar – Milk processing plant serving East India
- Mangolpuri, Delhi NCR – Fruits & Vegetables processing plant
- Ranchi, Jharkhand – Fruits & Vegetables processing plant
- Bengaluru, Karnataka – Fruits & Vegetables processing plant

3.2 Planned / Future Processing Plants

- Nagpur / Butibori, Maharashtra – Milk processing plant (Central India hub)
- Itola (near Vadodara), Gujarat – Fruits & Vegetables processing plant (West India)
- Kuppam, Andhra Pradesh – Fruits & Vegetables processing plant (South India)

3.3 Distribution Centres (Greenfield Candidates)

Candidate DC locations were selected from major logistics hubs across India to allow the optimization model to choose the most suitable ones:

- Delhi NCR
- Lucknow
- Patna
- Kolkata
- Guwahati
- Nagpur
- Bhopal
- Indore
- Ahmedabad
- Vadodara
- Jaipur
- Mumbai
- Pune
- Hyderabad
- Bengaluru
- Chennai
- Visakhapatnam
- Coimbatore

3.4 Market Regions

Demand was aggregated into five regions:

- M1 – NCR
- M2 – East India
- M3 – Central India
- M4 – West India
- M5 – South India

4. PROBLEM STATEMENT

Mother Dairy aims to expand its national footprint while ensuring that its supply chain remains feasible, cost-efficient, and resilient under cold chain constraints. The company must answer the following strategic questions:

1. Is the current network of processing plants sufficient to meet national demand under realistic cold chain and capacity constraints?
2. Where should new distribution centres be located to minimize cost and ensure feasibility?
3. How does the inclusion of future plants affect network feasibility and performance?
4. How resilient is the optimized network to operational disruptions?
5. Can improved logistics feasibility enable selective export opportunities as a secondary objective?

5. DATA AND MODELING ASSUMPTIONS

Due to the absence of proprietary operational data, the study adopts industry-based assumptions, which are standard in academic supply chain analysis.

- Transportation costs are distance-based and calculated using geographic coordinates (Haversine method).
- Cold chain feasibility is enforced using maximum allowable transport distance:
 - Milk: 500 km
 - Fruits & Vegetables: 800 km
- Demand must be fully satisfied (100% service level target).
- Inventory is managed using a Min–Max policy.
- Replenishment lead time is assumed to be 3 days.
- Demand variability is modelled using a normal distribution.

6. METHODOLOGY

6.1 Greenfield Analysis

Greenfield Analysis was conducted using a Python-based Mixed Integer Linear Programming (MILP) model. The model decides:

- Which distribution centres to open from candidate locations.
- How to route flows from plants → DCs → markets.

The objective is to minimize total cost, including transportation, handling, and fixed DC costs, subject to capacity and cold chain constraints.

Two scenarios were evaluated:

- Baseline scenario: Only existing processing plants available.
- Future scenario: Existing + future processing plants available.

6.2 Network Optimization

The optimized network incorporates:

- Binary decisions for opening DCs.
- Continuous flow decisions for product movement.
- Hard constraints on plant capacity, DC capacity, and cold chain distance.

6.3 Simulation Analysis

A time-based simulation was developed to evaluate dynamic performance over 365 days. The simulation tracks:

- Daily demand variability.
- Inventory levels.
- Stockouts.
- Service levels.

6.4 Risk and Disruption Analysis

A disruption scenario was simulated in which a major DC is unable to receive replenishment for 60 consecutive days. This scenario represents events such as infrastructure failure, transportation shutdowns, or natural disasters.

7. RESULTS AND DISCUSSION

7.1 Greenfield and Optimization Results

Baseline Network (Present Plants Only)

- Optimization status: Infeasible
- Average DC-to-market distance: ~721 km

The infeasibility indicates that the current network cannot satisfy national demand while respecting cold chain and capacity constraints. This finding is critical: the issue is not high cost, but structural insufficiency.

Future Network (Present + Future Plants)

- Optimization status: Optimal
- Selected DCs:
 - Lucknow
 - Patna
 - Indore
 - Hyderabad
- Average DC-to-market distance: ~389 km

7.2 Simulation Results

Under normal operating conditions:

- Service Level: 100%
- Temporary stockout events: 6
- All demand was ultimately fulfilled.

This indicates that the optimized network is operationally sound.

7.3 Risk Analysis Results

Under the disruption scenario:

- Service Level dropped to 86%
- Approximately 14% demand remained unmet.

8. STRATEGIC IMPLICATIONS AND EXPORT POTENTIAL

The primary objective of Mother Dairy's network expansion remains domestic demand fulfillment and farmer income protection, fully aligned with Indian government policy. However, the results reveal an important secondary insight.

By reducing transport distances and improving cold chain feasibility, the optimized network creates surplus-handling and resilience capacity. This opens the possibility for selective export of value-added and storable dairy products, such as:

- Skimmed Milk Powder
- Ghee and butter oil
- Cheese and processed dairy products

Exporting liquid milk is neither practical nor recommended. Export activity should be considered only after domestic demand is fully met and must comply with regulatory approvals.

Thus, exports are positioned as a secondary revenue-enhancing option, enabled by logistics efficiency rather than driven by it.

9. MANAGERIAL RECOMMENDATIONS

1. Mother Dairy must proceed with planned future processing plants, as the current network is structurally infeasible.
2. A regionalized distribution strategy with four strategically located DCs should be adopted.
3. Cold chain constraints must be embedded into all strategic network decisions.
4. Risk mitigation strategies, such as multi-sourcing and DC redundancy, should be strengthened.
5. Selective export of value-added dairy products may be explored during surplus periods, aligned with government policy and domestic food security.

10. CONCLUSION

This study demonstrates the value of prescriptive analytics in designing a robust cold chain network for perishable food distribution. Python-based optimization revealed that Mother Dairy's existing network is infeasible under realistic constraints, while planned future facilities are not only beneficial but essential. Simulation and risk analysis further validated network performance and resilience.

Beyond domestic optimization, improved feasibility enables strategic flexibility, including selective export opportunities. However, exports must remain a secondary objective, carefully aligned with policy and domestic priorities. Overall, the study highlights how data-driven network design can support sustainable growth, resilience, and strategic decision-making in India's dairy sector.

11. APPENDICES

Appendix A: Network Optimization Model Output

Objective

To evaluate the feasibility and performance of Mother Dairy's supply chain under two scenarios:

1. Baseline network (existing plants only)
2. Expanded network (existing + future plants)

The model determines optimal DC locations and transportation feasibility under cold-chain constraints.

Output

```
=== BASELINE (Present plants only) ===
```

```
Status: Infeasible
```

```
Total cost (₹/day): 1195455.13
```

```
Chosen DCs: ['CD3']
```

```
Avg DC->Market km: 721.1
```

```
=== FUTURE (Present + Future plants) ===
```

```
Status: Optimal
```

```
Total cost (₹/day): 1353758.99
```

```
Chosen DCs: ['CD2', 'CD3', 'CD8', 'CD14']
```

```
Avg DC->Market km: 388.8
```

Interpretation:

- The baseline network was found to be infeasible, indicating that existing plant infrastructure alone cannot satisfy national demand under capacity and cold-chain constraints.
- The inclusion of future plants enabled a feasible and optimized distribution network, significantly reducing average delivery distance and improving supply chain viability.

Appendix B: Inventory Simulation Model (Normal Conditions)

Purpose

To simulate inventory behaviour and service performance of the optimized network under demand variability.

```

import numpy as np
import pandas as pd

np.random.seed(42)

DAYS = 365
mean_demand = 100
std_dev = 15

min_inv = 300
max_inv = 600
lead_time = 3

inventory = max_inv
pipeline = []
stockouts = 0
served = 0
total_demand = 0

daily_inventory = []

for day in range(DAYS):
    demand = max(0, int(np.random.normal(mean_demand, std_dev)))
    total_demand += demand

    # Receive replenishment
    if pipeline and pipeline[0][0] == day:
        inventory += pipeline.pop(0)[1]

    # Fulfil demand
    if inventory >= demand:
        inventory -= demand
        served += demand
    else:
        served += inventory
        stockouts += (demand - inventory)
        inventory = 0

    # Reorder decision
    if inventory < min_inv:
        order_qty = max_inv - inventory
        pipeline.append((day + lead_time, order_qty))

    daily_inventory.append(inventory)

service_level = served / total_demand

print("Service Level:", round(service_level, 3))
print("Total Stockouts:", stockouts)

```

Output

```

Service Level: 1.0
Total Stockouts: 6

```

Interpretation

Under normal operating conditions:

- The system achieved a 100% service level.
- Six temporary stockout events occurred.
- However, all demand was ultimately fulfilled through timely replenishment.

Appendix C: Risk & Disruption Simulation Model

Purpose

To evaluate network resilience when replenishment is disrupted.

A disruption window was introduced between Day 120 and Day 180, representing operational shutdown or logistics failure at a distribution centre.

```
DISRUPTION_START = 120
DISRUPTION_END = 180

inventory = max_inv
pipeline = []
served = 0
total_demand = 0

for day in range(DAYS):
    demand = max(0, int(np.random.normal(mean_demand, std_dev)))
    total_demand += demand

    # Block replenishment during disruption
    if pipeline and pipeline[0][0] == day:
        if not (DISRUPTION_START <= day <= DISRUPTION_END):
            inventory += pipeline.pop(0)[1]
        else:
            pipeline.pop(0)

    if inventory >= demand:
        inventory -= demand
        served += demand
    else:
        served += inventory
        inventory = 0

    if inventory < min_inv and not (DISRUPTION_START <= day <= DISRUPTION_END):
        pipeline.append((day + lead_time, max_inv - inventory))

service_level_risk = served / total_demand
print("Service Level under Disruption:", round(service_level_risk, 3))
```

Output

Service Level under Disruption: 0.86

Interpretation

- Service level declined from 100% to 86% during disruption.
- Approximately 14% demand remained unmet.
- This demonstrates the supply chain's sensitivity to prolonged facility outages.

INDIVIDUAL CONTRIBUTIONS

K Tharun Kumar — Project Conceptualization & Problem Structuring

I was responsible for defining the overall project concept and structuring the analytical framework. The contribution began with identifying a relevant problem within India's cold chain logistics ecosystem, focusing on Mother Dairy's distribution expansion challenges. I developed the core research objectives, including evaluating network feasibility, facility expansion needs, and supply chain resilience. The analytical structure of the study Greenfield Analysis, Network Optimization, Simulation, and Risk Analysis was designed under my role. Additionally, I aligned the project with academic case-study standards and prescriptive analytics methodology. Early-stage team coordination, tool selection, and methodological planning were also facilitated. This foundational work ensured that the project remained strategically relevant, analytically structured, and academically rigorous, enabling subsequent modelling and simulation activities to proceed in a coherent and goal-oriented manner.

M Sireesha — Data Collection & Facility Mapping

I handled all data collection and geographic mapping activities. This included identifying existing and future Mother Dairy processing plants and categorizing them by operational status and product type. I mapped facility locations across India and compiled latitude-longitude coordinates required for distance-based modelling. Candidate greenfield distribution centre locations were also identified based on logistics feasibility and regional demand coverage. Additionally, India's demand markets were aggregated into five representative regions to support optimization modelling. All datasets were structured into standardized Excel sheets covering facilities, demand nodes, capacities, and candidate DCs. This data architecture enabled seamless integration with Python optimization models. Through accurate facility mapping and structured dataset preparation, I ensured that the analytical framework was grounded in realistic geographic and infrastructure conditions.

K Tejaswini Pranati — Network Optimization Modelling

I led the development of the Python-based network optimization model. The supply chain design problem was formulated as a Mixed Integer Linear Programming (MILP) model to determine optimal DC locations and shipment flows. Binary variables were used for facility selection, while continuous variables modelled transportation flows. Distance-based transport costs were computed using geographic coordinates. Capacity constraints for plants and DCs, along with cold-chain feasibility limits, were incorporated into the model. Two scenarios were evaluated: baseline (existing plants only) and future (including planned plants). The optimization results revealed that the baseline network was infeasible, while the expanded network produced an optimal solution. This modelling contribution provided the quantitative foundation for evaluating expansion feasibility and distribution efficiency.

S Aashiq Hussain — Simulation Modelling

I developed the time-based simulation model to evaluate operational performance under demand variability. The simulation was implemented in Python over a 365-day horizon. Daily demand fluctuations were modelled using probabilistic distributions, while inventory control was governed by a Min–Max replenishment policy. Lead time delays were incorporated through a shipment pipeline mechanism. The simulation tracked service levels, stockouts, and inventory levels. Results showed a 100% service level under normal operating conditions, validating the operational feasibility of the optimized network. This contribution extended the project beyond static optimization, offering dynamic insights into inventory stability and supply reliability.

P Anusha — Risk & Disruption Analysis

I conducted disruption modelling to evaluate supply chain resilience. A 60-day disruption scenario was simulated, during which replenishment to a distribution centre was blocked. The model assessed resulting impacts on inventory depletion and demand fulfilment. Simulation results indicated a service level decline to 86% under disruption conditions. This highlighted the network's sensitivity to prolonged facility outages and emphasized the importance of regional redundancy. The analysis strengthened the project's managerial relevance by incorporating risk preparedness and continuity planning into supply chain design evaluation.

S Niteswara Sai — Report Compilation & Strategic Analysis

I led report writing, results interpretation, and strategic integration. Responsibilities included compiling all analytical findings into a structured academic case study. Optimization, simulation, and risk results were translated into managerial insights and recommendations. The member also developed the strategic extension exploring selective export opportunities as a secondary objective enabled by improved logistics feasibility. Appendices, code documentation, and analytical explanations were prepared to ensure academic completeness. This contribution ensured clarity, coherence, and alignment between technical outputs and strategic conclusions.